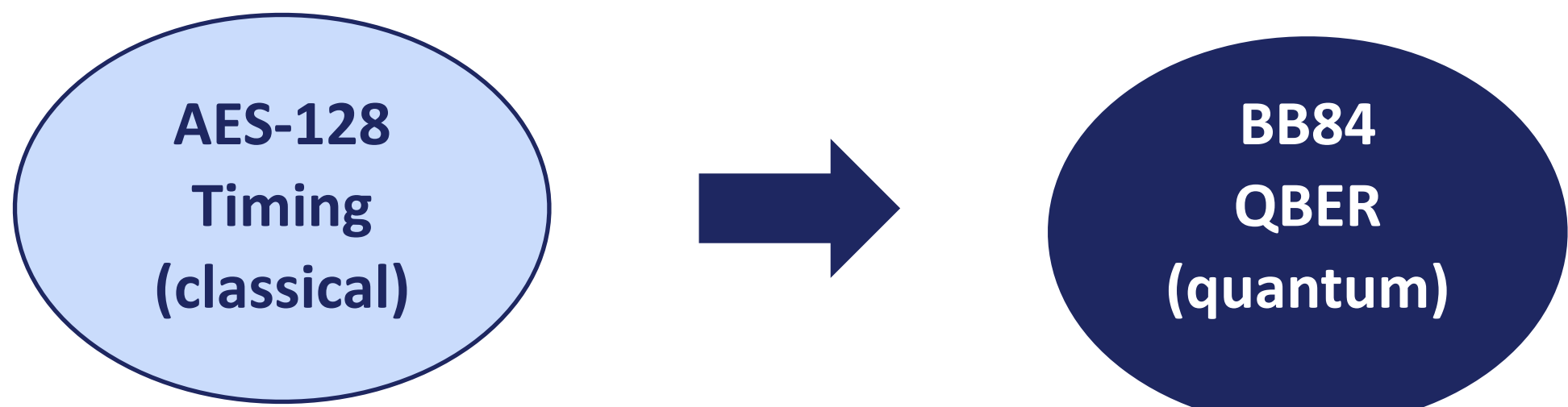


ABSTRACT

- Quantum Key Distribution (QKD) secures communication by exploiting quantum mechanics to detect eavesdropping during key exchange. This work presents QAURA, a lightweight anomaly detection framework for the BB84 protocol inspired by our previous AURA architecture for AES security.
- Using IBM Qiskit's Statevector simulator, BB84 sessions were generated under normal operation, intercept-resend attacks, and channel noise. Detection is performed using a statistical 3σ threshold, a Random Forest classifier, and a hybrid approach based on Quantum Bit Error Rate (QBER).
- Experimental results demonstrate accurate detection of eavesdropping with low computational overhead, showing that the AURA detection framework can be effectively adapted from classical embedded security to quantum communication.

BACKGROUND

- Prior work** (AURA) built a lightweight real-time anomaly detector for AES-128 encryption timing side-channels on embedded systems (CPU and Xilinx PYNQ-Z1 FPGA), combining a statistical threshold rule, a Random Forest classifier, and a hybrid mode.
- The gap:** AES-128 timing side-channels are a property of classical hardware execution. There is no meaningful “quantum computing” analog of a timing attack, since AES is not executed on a quantum processor — so AURA cannot be ported to quantum computing literally.
- Our approach:** instead of forcing that translation, we carry forward AURA's actual contribution — the detection architecture itself — into a domain that is genuinely quantum: eavesdropping detection on a BB84 quantum key distribution (QKD) channel.
- Why BB84:** any eavesdropper intercepting photons on a quantum channel unavoidably disturbs the quantum state, introducing a measurable rise in the Quantum Bit Error Rate (QBER) — the direct quantum counterpart of a timing anomaly.



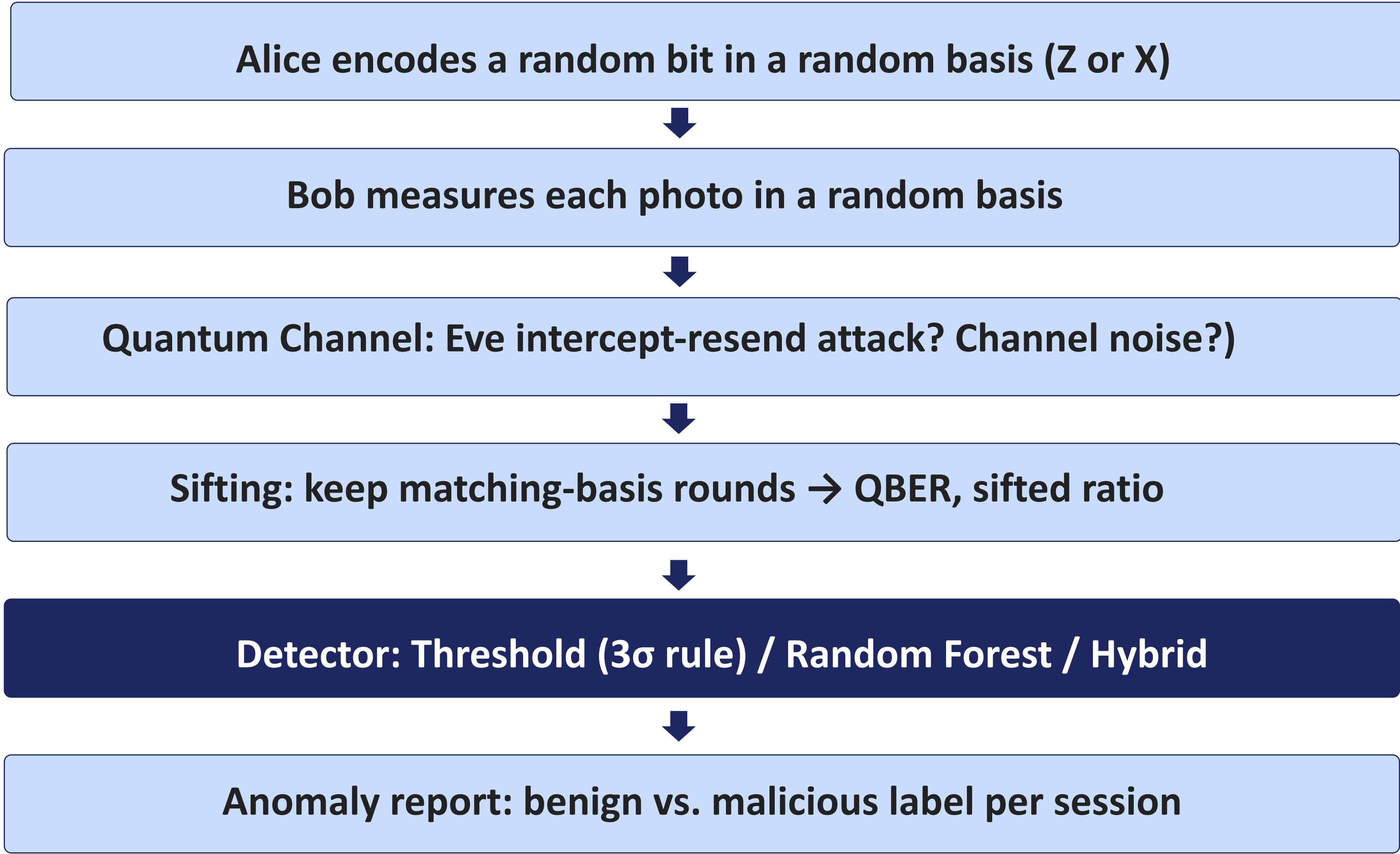
same detection architecture, applied to a genuinely quantum signal

RESOURCES

- Chinnasami, N., Stable-Smith, R., & Karakchi, R. (2025).** *ML-Enhanced AES Anomaly Detection for Real-Time Embedded Security*. arXiv:2507.04197.
- Bennett, C. H., & Brassard, G. (1984).** *Quantum Cryptography: Public Key Distribution and Coin Tossing*. Proceedings of IEEE International Conference on Computers, Systems and Signal Processing, Bangalore, India, pp. 175–179.

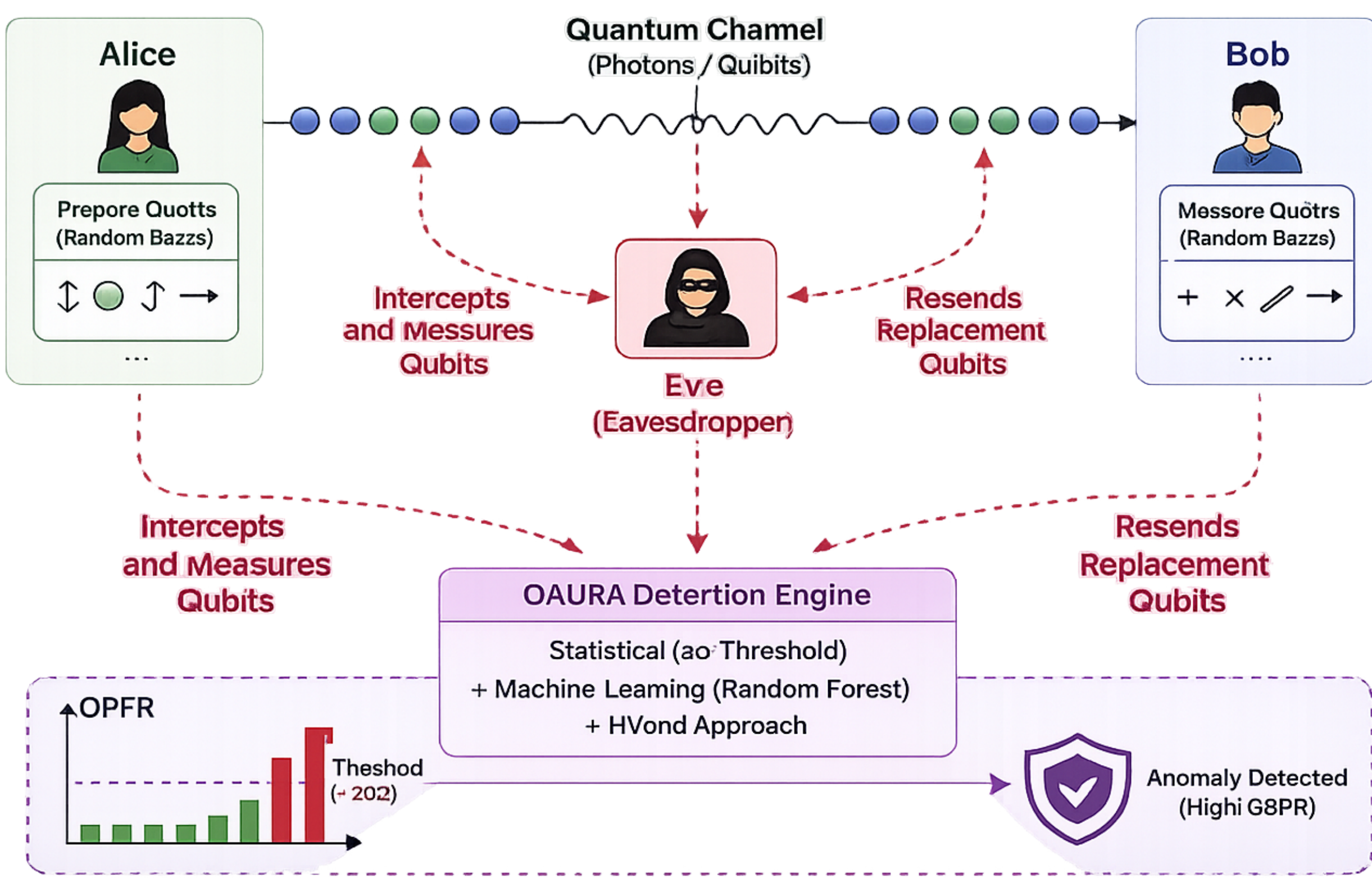
METHODOLOGY

BB84 Session Pipeline



THREATS TO QUANTUM COMMUNICATION

- Intercept-Resend Attack:** An eavesdropper (Eve) measures transmitted qubits and resends replacements, introducing detectable errors.
- Channel Noise:** Environmental interference and hardware imperfections increase the Quantum Bit Error Rate (QBER), potentially masking attacks.
- Basis Mismatch:** When Alice and Bob use different measurement bases, the corresponding bits are discarded during key reconciliation.
- Stealthy Eavesdropping:** Partial interception strategies attempt to minimize QBER while still extracting information from the key exchange.



CONCLUSION & FUTURE WORK

- AURA's threshold / ML / hybrid detection architecture generalizes cleanly from a classical side-channel (AES timing) to a genuinely quantum one (BB84 QBER).
- Future work: validate on real IBM Quantum hardware via Qiskit Runtime; extend to E91 / B92 protocols for cross-protocol benchmarking, paralleling AURA's cross-modality video benchmarking.

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RESULTS

RESULTS: DETECTION PERFORMANCE

- 400 simulated BB84 sessions (150 qubits each): 275 clean, 64 noise-spike, 61 Eve intercept-resend attacks. Eve's attack produces a QBER of ~25%, matching BB84 theory's predicted intercept-resend error rate exactly.

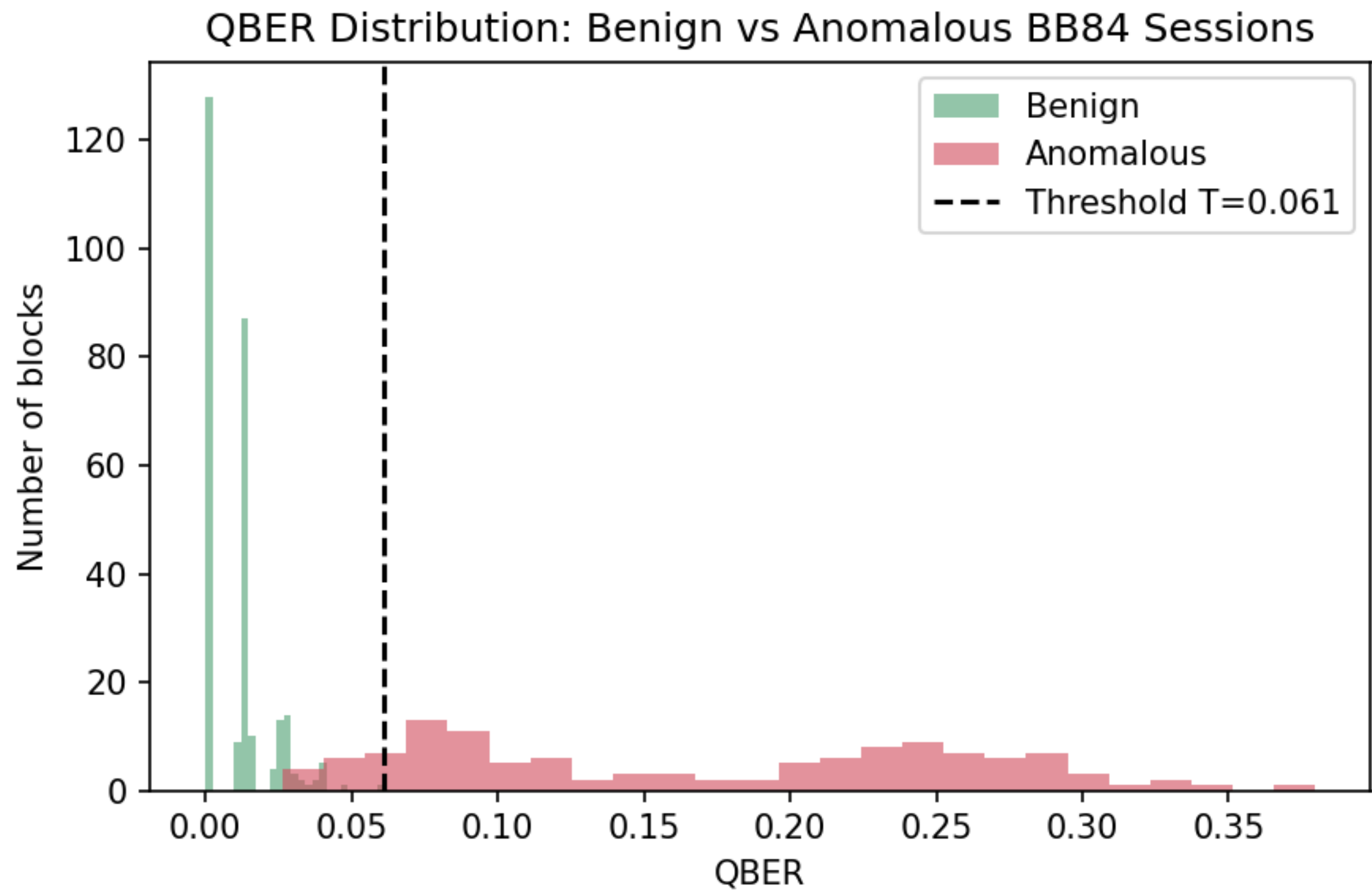


Figure 1: QBER cleanly separates benign from anomalous sessions.

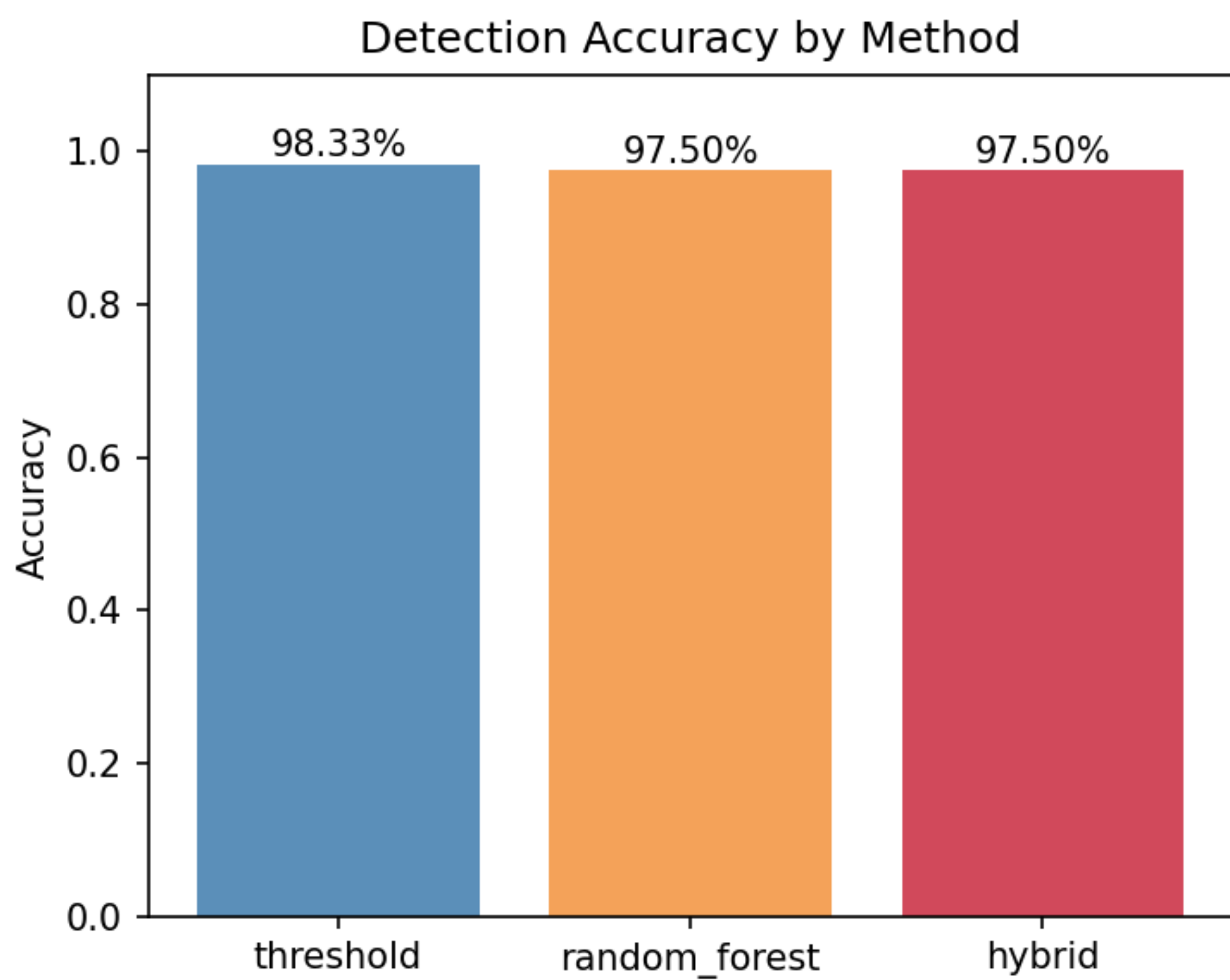


Figure 2: Threshold 98.3% | Random Forest 97.5% | Hybrid 97.5% accuracy

Threshold		Random Forest		Hybrid	
True Benign	83	81	81	81	81
True Anomaly	1	0	0	0	0
Pred Benign	83	81	81	81	81
Pred Anomaly	1	0	0	0	0

Figure 3: Confusion matrices — Threshold, Random Forest, and Hybrid detectors.

ACKNOWLEDGMENT

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